

Anish Ghosh

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EDUCATION

- **University of Maryland, Baltimore County** Baltimore, Maryland
Ph.D. Applied Mathematics 2025 - present
- **Drexel University** Philadelphia, Pennsylvania
MSc Mathematics 2023 - 2025
- **Constructor University** Bremen, Germany
BSc Computer Science 2019 - 2022

RESEARCH EXPERIENCE

- **University of Maryland, Baltimore County (UMBC)** Maryland, USA
Ph.D. Student, Research Assistant — Applied Mathematics Aug 2025 – Present
 - Research Assistant since Jun 2026.
 - Advised by Dr. Mohammadhossein Mohammadisiahroudi and Dr. Matthias Gobbert.
 - Researching hybrid quantum algorithms for PDE-constrained optimization.
 - Developing operator-learning methods to precondition stiff matrices arising from PDEs.
- **Fraunhofer FHR** Bonn, Germany
Research Assistant - Algorithm and Software Development Jun 2022 - Dec 2022
 - Developed and implemented software components for the GESTRA project, focusing on system reliability and performance optimization.
 - Established a communication network using MODBUS/TCP protocol for multiple Pqube3 devices.
 - Contributed to monitoring for the GESTRA radar project, ensuring seamless integration of subsystems.
 - Utilized Qt Framework, C++, Linux for implementation.
- **Fraunhofer MEVIS** Bremen, Germany
Bachelor Thesis Mar 2022 - Dec 2022
 - Conducted comprehensive literature research on Generative Adversarial Networks (GAN), with a focus on transformer-based architectures and advanced image generation techniques.
 - Worked extensively on developing a novel metric for comparing the quality of GAN-generated images, building upon the Fréchet Inception Distance (FID) metric and leveraging Transformer Architecture to improve comparison precision.
 - Implemented the metric using PyTorch, and NumPy, optimizing it for scalability and efficient processing of large image datasets.
 - Designed and developed a Transformer Network for comparing and analyzing generated images, nearing completion to provide deeper insights into model performance and output quality.
- **3d.aero** Hamburg, Germany
Working Student - Algorithm and Software Development Sep 2021 - Feb 2022
 - Researched and implemented 2D/3D image processing and computer vision algorithms in C# for industrial applications, improving image detection and processing accuracy.
 - Designed and implemented user interface components using XAML to improve usability and user experience.

- Worked independently to address challenging technical problems, ensuring efficient and reliable solutions.

• **Fraunhofer IPA**

Stuttgart, Germany

Internship - Software and Algorithm Development

Jun 2021 - Aug 2021

- Developed automatic grasp generation algorithms for bin-picking systems, increasing operational accuracy.
- Analyzed and improved existing algorithms for bin-picking systems to increase operational accuracy.
- Developed and implemented an algorithm to convert .stl files to .pcd format, optimizing point cloud data for 3D processing.
- Worked with libraries such as PCL, OpenCV, and HDF5 to streamline algorithm development.
- Reduced time complexity of algorithms by identifying and optimizing performance bottlenecks.
- Gained hands-on experience in integrating advanced algorithms into industrial automation systems.

TEACHING EXPERIENCE

• **University of Maryland Baltimore County**

Baltimore County, Maryland

Teaching Assistant - Mathematics Department

Aug 2025 - Jun 2026

- Delivered tutorials and practice materials for students in Calculus II, Multivariable Calculus.
- Graded quizzes and exams, providing constructive feedback to improve student performance.

• **Drexel University**

Philadelphia, Pennsylvania

Teaching Assistant - Mathematics Department

Jan 2023 - Jun 2025

- Delivered tutorials and personalized learning sessions for students in Multivariable Calculus, Probability, and Intro to Analysis.
- Graded assignments and exams, providing constructive feedback to improve student performance.
- Assisted students with a wide range of mathematics courses at the Math Resource Center, providing support tailored to their specific questions and challenges.

• **Constructor University**

Bremen, Germany

Teaching Assistant - Stochastic Methods

Sep 2022 - Dec 2022

- Topics: Binomial tree models, Discrete Brownian paths, stochastic integrals and ODEs, Ito's Lemma, Monte-Carlo methods, finite differences solutions, the Black-Scholes equation, and an introduction to time series analysis, parameter estimation, and calibration.
- Addressed student questions on these topics during office hours and tutorials.
- Graded and automated a portion of the grading for assignments using Python scripts, reducing feedback turnaround time.

• **Constructor University**

Bremen, Germany

Teaching Assistant - Applied Dynamical Systems

Sep 2021 - Dec 2021

- Topics: ODE solvers (forward, backward euler methods, etc.), Logistic Map, Simple Model of Biochemical Switches and Oscillators, Chua's Circuit, Cellular Automata and Belousov-Zhabotinsky reaction
- Delivered tutorials and addressed student questions on these topics.
- Graded assignments and reports involving Python and Mathematica, ensuring detailed feedback and improvements

• **Constructor University**

Bremen, Germany

Teaching Assistant - Human Computer Interaction

Sep 2021 - Dec 2021

- Guided students in face detection, optical flow estimation, and GUI development using OpenCV, C++, CMake, and NodeJS.

- Provided detailed assignment feedback, enhancing students' practical understanding of computer vision concepts.

• Constructor University

Bremen, Germany

Teaching Assistant - Human Computer Interaction

Feb 2021 - May 2021

- Wrote lecture slides and guided students in implementing face detection, optical flow estimation, and GUI.
- Graded assignments, focusing on computer vision tasks using OpenCV and C++.

CURRENT RESEARCH PROJECTS

- **Quantum Accelerated Adjoint Method for PDE-Constrained Optimization:** Hybrid quantum-classical framework replacing the classical adjoint linear solve with a Quantum Linear System Algorithm, achieving exponential speedup with respect to the number of state variables.

UNDERGRADUATE PROJECTS (2020 – 2022)

- **GESTRA Radar System:** Played a key role in optimizing communication protocols and controlling the radar system for the GESTRA project. Developed software components focused on improving hardware communication. Implemented a MODBUS/TCP protocol for multiple Pqube3 devices and contributed to the system's control. This improved communication reliability and data transmission rates.
- **Rivet Detection and Measurement (3d.aero):** Designed and implemented an image filter using MSBF algorithm to detect rivets and edges based on input from a line sensor, improving detection accuracy and consistency across various material types.
- **Bushing Measurement (3d.aero):** Developed a stereo camera-based filter for measuring the radius and height of bushings. Integrated the filter with a look-up table for efficient identification, reducing error rates in detecting faulty bushings.
- **Bin Picking (Fraunhofer IPA):** Developed an automatic grasp generation algorithm for parallel jaw and suction cup grippers, integrated into bin-picking software. Improved picking accuracy and object handling efficiency in industrial applications.
- **Viola Jones Face Detector:** Implemented a real-time Viola-Jones face detection system using OpenCV and cascade classifiers. Developed with C++, CMake, and Xcode, the system utilized pre-trained classifiers and measured FPS, improving detection efficiency.
- **Xilinx Synthesis:** Designed and implemented an 8-bit adder, AND, and XOR gates using VHDL on Xilinx. Synthesized the circuits, validating functionality and performance for further hardware deployment.
- **FPGA Programming:** Programmed a seven-segment display on a Spartan FPGA board.

SKILLS

- **Programming Languages:** Python, C/C++, C#, .NET Framework, JavaScript, MATLAB, Julia, Angular, NodeJS, Java, SQL, HTML/CSS, PHP, Lua, Flutter/Dart, Latex
- **Hardware Description Languages:** VHDL, Assembly
- **Development Environments:** Visual Studio, Android Studio, Xcode
- **Libraries and Tools:** Qiskit, OpenCV, NumPy, SciPy, Pandas, PyTorch, PCL, Open3D, Keras
- **Cloud & DevOps:** Google Firebase, Docker, Jenkins, Cloud Compare
- **Version Control and Collaboration:** Git, GitLab, BitBucket, SourceTree
- **Software Testing and Debugging:** Valgrind, GDB, NUnit

REFERENCES

- **Dr. Matthias Gobbert**

Professor

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- **Dr. Mohammadhossein Mohammadisiahroudi**

Assistant Professor

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